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## 6.1 CALIBRATION EQUIPMENT AND CONFIGURATION

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### INITIAL SETUP:

**Tools and Special Tools**

Site Calibration Kit (Drawing 920-00636)

**References****Materials/Parts**

None

**Equipment Condition**

Configuration WP's Completed

Equipment Powered On

Generator or Shore power applied

**Personnel Required**

Technician (2)

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### GENERAL INFORMATION

Before calibration, it is important to verify that vegetation and/or snow has been cleared within tolerances. See Maintenance Procedure C-3 Maintenance of Critical Areas.

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### Calibration Equipment Configuration

#### FTM Assembly and Setup

**NOTE**

Always start the FTM application from the MI, by selecting "Start FTM" in the MI Tools menu. Starting FTM from Windows Explorer will result in a failure for the MI and FTM to communicate correctly.

1. Place the Theodolite and other assembly components on a flat table near the MI.

**NOTE**

Theodolite will not transmit test data if unit is not properly leveled.

2. For dual rack system ensure System is functioning from RACK "A" by verifying light is on for RACK "A" on the Rack Auto Switch Panel.
3. If system is not functioning from RACK "A", click "Switch Racks" from the Controls pulldown menu on the RCU.
4. Pull free end of CAT5 cable to interface panel at port labeled FTM.
5. Attach cable strain relief to TLS interface Strain relief hook.
6. Plug the power USB into the Theodolite
7. Connect Theodolite serial converter to Theodolite.

8. Connect CAT5 cable to Theodolite serial converter.
9. Power on the Theodolite.
10. Level Theodolite IAW components manufacturer's instructions.
11. Select Flight Test Monitor (FTM) from "Start FTM" in the tools menu on MI.
12. Select "RS-232" from the FTM Theodolite menu.
13. Select "COM2" and "9600 Baud" from the dialog box that opens.
14. Click "OK" to save and exit back to the main FTM window.
15. Click the green "P" (Ping Theodolite) button.
16. Verify elevation and horizontal data is displayed in the FTM ping test window.
  - a. Check the cable connections from the MIU to Theodolite and serial port settings on the MIU if data is not displayed.
17. Close FTM ping test window.

**END OF TASK**

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## 6.2 CALIBRATION

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### INITIAL SETUP:

**Tools and Special Tools**

Calibration Kit

**References****Materials/Parts**

None

**Equipment Condition**Calibration Equipment and Configuration WP  
Completed

Equipment Powered On

Generator or Shore power applied

**Personnel Required**

Technician (2)

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### GENERAL INFORMATION

The Calibration section includes:

- System configuration
- Pre-Flight calibration
- Pre-flight planning
- Calibration profiles
- Calibration file generation
- Calibration validation verification
- Alternative calibration procedure
- Calibration quality
- Adjustment/verification
- Settings validation.

Verify coordination with airport authorities for access to runway. The best most accurate result is achieved from the centerline of the runway.

#### WARNING

Hazardous Radio Frequency (RF) radiation is present during parts of System Calibration. The area near the antenna shall be clear of personnel, munitions, fuels, structures, obstacles, and equipment for at least 1 meter around the antenna when radiating. Failure to comply may result in death or injury to personnel and/or damage to equipment.

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### Calibration Procedures

#### Base Station Configuration

1. With the Maintenance Interface window open select "System Setup."
2. Select Settings.
3. The Box labeled Data Recording ensure the following boxes are checked:
  - a. Record Track Ram files Guidance
  - b. Record Raw files

- c. Record GTU files
4. The Box Labeled Data Recording enter the First Character of the Raw File Name, the default location is "x".
5. Restart the MI software to effect changes.

## Pre-Flight Calibration

**Perform the procedure steps of Appendix A to complete the initial alignment of the TLS localizer and glide slope.**

## Pre-Flight Planning

### NOTE

- Brief the TLS test operator and pilot prior to staging on the runway.
  - Utilize TLS CTF Checklist to record data and archive information.
  - The Preflight brief is based on a 3.0° approach and an intercept altitude of 1600' AGL. Adjust the preflight briefing as necessary to adjust for specific site criteria and approach data being used for calibration.
1. TLS CTF profiles:
    - a. Two at constant altitude (level crossing) on the approach centerline at the glide slope intercept elevation (if known) or 1600' AGL from 10 nm. There will be no TLS guidance during this approach, so the pilot only needs to fly the approach to the best of their abilities.
    - b. One profile from 10 nm, starting at altitude 1600', near the glide slope intercept (approx. 5 nm from threshold for a 3.0° approach), the pilot descends on glide path to threshold (this approach could be full stop)
  2. TLS CTF Pre-flight brief for Operator:

### WARNING

- The TLS operator(s) SHALL brief the pilot(s) about general concepts of TLS operation, and review how the acquisition - track aspects of the system work.
  - If the pilot should fly the initial calibration profile visually and if he receives needle indications during the initial calibration profile, the needles should be ignored.
- a. Coordinate radio communications between the FTM operator and the TLS operator on a ground-to-ground radio frequency that will not interfere with ground-to-air communications.
  - b. All profiles need to be near the approach centerline and heading, and each at its respective constant altitudes (with the exception of the last approach). If the theodolite location at the edge of the runway is to be used for calibration, the pilot(s) will be asked to fly a course to that edge of the runway - not to runway centerline.

- c. The course shall be maintained until the TLS operator requests the pilot to break off the approach and set up for the next profile.
  - d. Each calibration profile will be considered complete when the theodolite operator has successfully tracked the aircraft, and base operator has recorded the FTM file from 1° elevation to at least 6° or 2° above the desired glide slope angle which is typically 3°.
  - e. For each profile, verify and document the quality of collected data in CTF checklist before proceeding. Review the items in the following verification list:
    - i. Did the pilot adequately fly the desired profile?
    - ii. Did the theodolite operator successfully track the aircraft?
    - iii. Was the .dif file saved, and does it have measurements?
    - iv. Was the .raw file saved, and does it have measurements?
3. TLS CTF Pre-flight brief for Pilot:
- a. The Transponder Landing System (TLS) is a ground-based instrument navigational aid that provides precise lateral and vertical guidance for CAT I approaches. The TLS operates as an Instrument Landing System but can be established in austere environments where a conventional ILS could not be. Like that of a PAR controller the TLS is also capable of providing Precision Approach Radar guidance.
  - b. The Pilot is responsible for all safety decisions during the calibration profiles.
  - c. If an approach must be aborted due to safety reasons, the pilot will repeat the approach.
  - d. Establish callsigns between pilot and TLS operator, i.e., "28H, TLS Base."
  - e. Coordinate engine start time with TLS operator.
  - f. The pilot will request and receive a transponder code from ATC and relay this code to the TLS operator.
  - g. The calibration aircraft must be equipped with a transponder, ILS receiver, air-to-ground radio. Landing lights are nice to have.
  - h. The pilot will turn on the landing lights during each approach to facilitate tracking with the theodolite.
  - i. The pilot shall be briefed to ignore any instrument indications during level crossing calibration profiles
  - j. Radio communications between the base station and the calibration aircraft shall be maintained at a radio frequency to be agreed to prior to the profiles.

| TLS Calibration Test Flight (CTF) MISSION BRIEF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose:</b><br>The purpose of the CTF is to calibrate the TLS using a piloted aircraft, TLS operator and a theodolite operator positioned at or near the runway centerline. The theodolite operator will visually track the piloted aircraft that will be flying a predetermined flight profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CTF TEST PLAN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>1. Three data collection profiles are required for the calibration data set:</p> <p>a. Two at constant altitude (level crossing) on the approach centerline at the glide slope intercept elevation (if known) or 1600' AGL from 10nm. There will be no TLS guidance during this approach, so the pilot only needs to fly the approach to the best of their abilities.</p> <p>b. One profile from 8 NM, starting at altitude 1600', near the glide slope intercept (approx. 5 NM from threshold for a 3.0° approach), the pilot descends on glide path to threshold (this approach could be to full stop).</p> <p>2. If low visibility is forecasted and a helicopter is available follow "Low visibility specila instructions" in WP Calibration Procedure, TM 1-XXXX-XXX-13</p> <p>3. All profiles (except for localizer crossings) need to be near the approach centerline and heading, and each at its respective constant altitudes (with the exception of the last approach). If the theodolite location at the edge of the runway is to be used for calibration, the pilot(s) will be asked to fly a course to that edge of the runway - not to runway centerline.</p> <p>4. The course shall be maintained until the TLS operator requests the pilot to break off the approach and set up for the next profile.</p> <p>5. Each calibration profile will be considered complete when the theodolite operator has successfully tracked the aircraft, and base operator has recorded the FTM file from 1 degrees elevation to at least 5 degrees or two degrees above the desired glide slope angle which is typically 3 degrees.</p> | <p>6. Radio communications between the base station and the calibration aircraft shall be maintained at a radio frequency to be agreed to prior to the profiles.</p> <p>7. The pilot is responsible for all safety decisions during the calibration profiles. If an approach has to be aborted due to safety reasons, the pilot will repeat the approach.</p> <p>8. The pilot will request and receive an transponder ID code from ATC and relay this code to the TLS operator.</p> <p>9. Coordinate radio communications between the FTM operator and the TLS operator on a ground-to-ground radio frequency that will not interfere with ground-to-air communications.</p> <p>10. The calibration aircraft must be equipped with a transponder, ILS receiver, air-to-ground radio, and landing lights. The pilot will turn on the landing lights during each approach to facilitate tracking with the theodolite.</p> <p>11. The pilot shall be briefed to ignore any needle indications while the TLS is operating prior to calibration process being completed.</p> |
| CTF GENRAL INFORMATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Aircraft Type: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | TLS Operator: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Aircraft Number: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | THEO. Operator: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Pilot: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DATE: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Pilot: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Airfield I.D. : _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| COMM FREQ: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | XPDR Code: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| TLS Calibration Completion<br>SIGN: _____ DATE: _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| TLS Calibration Test Flight (CTF)                                       |  |                                                                                                                                                                                    |
|-------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE OF CTF                                                          |  | ACFT TYPE:      DATE:                                                                                                                                                              |
| PILOT                                                                   |  | TLS OPERATOR                                                                                                                                                                       |
| SYMBOLS: v = SATISFACTORY, X = DEFICIENCY                               |  |                                                                                                                                                                                    |
| PRIOR TO CTF                                                            |  | * If unable to meet CTF criteria due to terrain limitations or site obstructions contact ANPC Field Service Representative for support in developing a site specific CTF profile.* |
| 1. Coordinate runway access.                                            |  |                                                                                                                                                                                    |
| 2. Pre-Flight plan briefed.                                             |  |                                                                                                                                                                                    |
| 3. Setup/Ping Theodolite                                                |  |                                                                                                                                                                                    |
| 4. Position and Level Theodolite.                                       |  |                                                                                                                                                                                    |
| 5. Establish Comm Frequency.                                            |  |                                                                                                                                                                                    |
|                                                                         |  | GLIDE PATH FLIGHT PROFILE #1                                                                                                                                                       |
|                                                                         |  | 1. Level at 1600' AGL.                                                                                                                                                             |
| CONSTANT ALT. FLIGHT PROFILE #1                                         |  | 2. Approach from 8nm.                                                                                                                                                              |
| 1. Level at 1600' AGL.                                                  |  | 3. Start descent (approx. 5nm from threshold)                                                                                                                                      |
| 2. Approached from 10nm.                                                |  | 4. Maintain a 3.0 degree glide path.                                                                                                                                               |
| 3. Maintain Level approach.                                             |  | 5. Verify with TLS operator.                                                                                                                                                       |
| 4. Verify with TLS Operator.                                            |  | NOTES:                                                                                                                                                                             |
| NOTES:                                                                  |  |                                                                                                                                                                                    |
|                                                                         |  | GLIDE PATH FLIGHT PROFILE #2                                                                                                                                                       |
| CONSTANT ALT. FLIGHT PROFILE #2                                         |  | 1. Level at 1600' AGL.                                                                                                                                                             |
| 1. Level at 1600' AGL                                                   |  | 2. Approach from 8nm.                                                                                                                                                              |
| 2. Approached from 10nm.                                                |  | 3. Start descent (approx. 5nm from threshold)                                                                                                                                      |
| 3. Maintain Level approach                                              |  | 4. Maintain a 3.0 degree glide path.                                                                                                                                               |
| 4. Verify with TLS Operator.                                            |  | 5. Verify with TLS operator.                                                                                                                                                       |
| NOTES:                                                                  |  | NOTES:                                                                                                                                                                             |
|                                                                         |  |                                                                                                                                                                                    |
| CTF Quality Checks                                                      |  | Additional Flight Profiles Needed                                                                                                                                                  |
| NOTE: If additional profiles required, continue on a new CTF checklist. |  |                                                                                                                                                                                    |
| COMMENTS:                                                               |  |                                                                                                                                                                                    |

**NOTE**

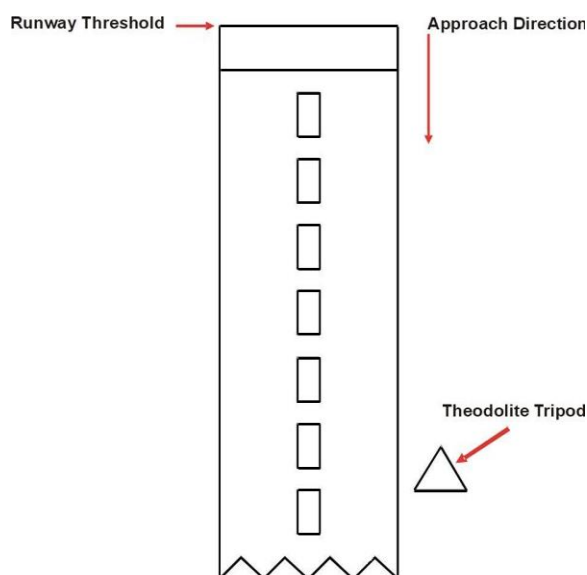
If the visibility conditions are expected to be poor and prohibit being able to see the aircraft at 8 to 10 nm, then the low visibility procedure should be used as follows.

**Perform the following Quality Checks**

1. Theodolite batteries charged.
2. Hand-held radios charged.
3. Theodolite can be test pinged before briefing pilot.
4. Set the RCU to be used with FTM on the MIU settings screen.
5. Pilot briefed on the required approaches.
6. Pilot knows to ask for a transponder code assignment.
7. Pilot knows what ILS frequency to tune.
8. Pilot knows to turn on landing lights.
9. Pilot knows the first two approaches are visual, and cannot trust Loc.
10. Approach.dat has an offset option if not using the centerline.
11. Pilot briefed on offset theodolite approach if used.
12. Pilot briefed on verification low approach only if theodolite is on the runway.

**Theodolite Setup for Calibration or Flight Inspection****NOTE**

- Use this setup procedure for both the calibration and the flight check procedures.
  - Before you begin the procedure, contact Air Traffic Control (ATC) to coordinate a time when it is safe to work next to the runway.
  - Before proceeding ensure FTM Theodolite Position Survey is complete, refer to Equipment Survey WP.
  - Theodolite serial data converter is connected and setup, feed CAT5 cable from reel to surveyed location.
1. Carry the CAT5 cable reel from the TLS shelter to the pre-surveyed location.
  2. Position Theodolite at the pre-surveyed location, reference Precision Equipment Survey worksheet.
  3. Attach Theodolite to the tripod using the screw attached to the tripod thumb screw on base of Theodolite.
  4. Center the tripod over the designated position with one leg pointing towards the approach direction (See Figure 2-1).



**Figure 2-1. Theodolite position.**

**NOTE**

- Theodolite position can be on runway centerline if permitted by ATC and airport operations.
  - If the location is offset from centerline an offset theodolite approach must be configured in the approach.dat file.
5. For centerline of runway approaches:
    - a. Align theodolite so eye piece will be located on the glide slope and in-line with the centerline of runway.
  6. For offset approaches:
    - a. Align theodolite so eyepiece will be located parallel to centerline and on the glide slope.
    - b. Use the pre-surveyed back sight to ensure eyepiece is parallel with runway centerline.
  7. Adjust height of Theodolite eye piece the same height as the person performing calibration process.
  8. Level theodolite IAW manufacturer's instructions.
  9. Zero the theodolite azimuth on the previously marked backsight, refer to Equipment Survey WP.
  10. Ensure knobs for adjustment on theodolite are facing away from operator to aid in tracking aircraft. If needed rotate theodolite 180° to position.
  11. Theodolite screen should indicate 270° for Z-A axis.

12. Ensure enough vertical vernier adjustment to reach at least 5° up elevation, and that horizontal vernier range of adjustment is approximately centered on the surveyed backsight so that aircraft tracking from left to right can be accommodated.
13. Connect FTM Extender to theodolite and CAT5 cable.
14. Verify handheld radio is tuned to the correct frequency and notify the TLS base station technician that the Theodolite is ready.
15. Perform the following Theodolite Setup Quality Checks.
  - a. Theodolite properly leveled at surveyed location.
  - b. FTM Cable correctly routed with strain relief.
  - c. Verify pings from FTM software.
  - d. Theodolite azimuth zeroed on the back-site/centerline (the direction that the planes approach)
  - e. Theodolite knobs adjusted to allow access to service volume elevation 0 to 7°, azimuth  $\pm 3^\circ$
  - f. Handheld ground-to-air radio is tuned to the correct frequency.

## FTM Setup

### NOTE

The following procedure will be completed at TLS base station using the MI.

1. Exit FTM if it is running.
2. Select “Start FTM” from the Tools pull down menu.
3. Select “RS-232” from the Flight Track Monitor Window’s Theodolite pull down menu.
4. Select glide slope location area.
5. Input the distance from runway threshold to the Theodolite position.

### NOTE

- The inputted measured distance should be negative.
  - Record number to the nearest tenth.
6. Click “OK” button to close “RS-232” setup window.
  7. Verify that Theodolite is setup.
  8. Select “Ping Connection” from the Theodolite pulldown menu.
  9. Verify “Ping” is shown on the FTM display.
  10. Select “L” button from the FTM window.

11. Enter Preflight information in the log entry form for weather, aircraft, pilot, type and purpose of flight.
12. Proceed to Calibration profiles.

## Calibration Profiles

Coordinate the time of aircraft flight so that no aircraft flight time is lost while waiting for the FTM- Base communication links to be established and tested.

Use the following Checklist in conjunction with performing the Calibration Profile steps:

1. ☐ Theodolite is in position and level
2. ☐ Handheld comms with theodolite operator is good to go
3. ☐ Select the FTM tool from the MIU Tool menu, press “L,” and then “Ok.”
4. ☐ Select Test Mode from the MIU System menu option
5. ☐ Enter the transponder code on the MIU test mode screen
6. ☐ Press “Acquire ILS” when the pilot reports turning inbound
7. ☐ Keep the theodolite operator updated on the aircraft position
8. ☐ After the theodolite operator reports he is tracking, press record on the FTM tool
9. ☐ At the end of the approach, press stop on the FTM tool and save the file
10. ☐ Follow the Syscal SW tool instructions to calibrate the TLS
11. ☐ Save the new cal.dat to the MIU, Base1, and Base2
12. ☐ 1<sup>st</sup> level approach successful
13. ☐ 2<sup>nd</sup> level approach successful
14. ☐ OPTION: Use the Mode-C calibration option if theodolite data is missing or challenged by visibility
15. ☐ Dif files generated and saved in the correct location
16. ☐ Syscal plots were normal and saved
17. ☐ New cal.dat file generated in Syscal is saved correctly
18. ☐ Verification approach completed, and signal verified by the pilot

1. Switch the front panel power switch of the GTU(s) to off.
2. From the FTM window Click on the "L" icon to open a new FTM document.
3. Enter the Preflight information.

#### NOTE

The Preflight information can include specifics about the weather, aircraft, pilot, type, and purpose of flight, etc.

4. 5. Click "OK".
5. Instruct aircraft to make a procedure turn onto the first approach (level at minimum altitude) at 10 NM, and to inform the TLS operator when the aircraft is starting the level run.
6. From the MI Tools menu, select "Test Mode" (See Figure 2-2).



**Figure 2-2. Test Mode Selection**

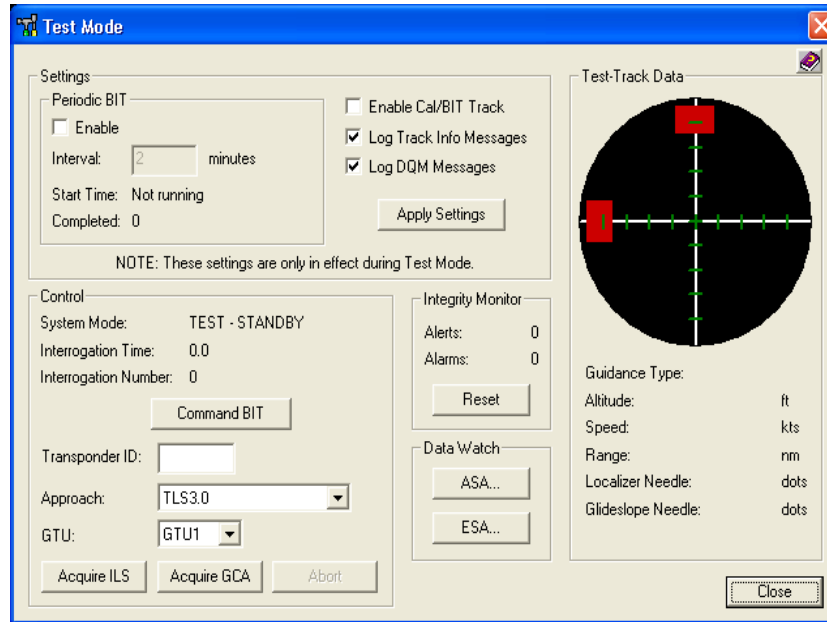
#### NOTE

- The Offset approach for calibration is the offset from runway centerline to the edge of the runway.
- The approach selected will be determined on what approach is being used and if a site-specific approach was created in the approach.dat application.

7. From the Test Mode window, select the approach profile being used for the calibration profiles
8. Enter the Transponder ID that will be utilized for the calibration profiles (See Figure 2-3).
9. On the MI in Test Mode, after the pilot indicates that the approach has started, with aircraft approximately aligned with the runway and lined up on the calibration approach, heading

inbound at the first altitude, and theodolite operator searching for aircraft, click the "Acquire ILS" button (See Figure 2-3).

10. Once aircraft is inbound, select "Acquire ILS".



**Figure 2-3. Test Mode Screen**

11. Verify "TRACK LOC" is displayed for System Mode in the Test Mode window.

#### NOTE

- Refer to theodolite manufacturer's instructions for detailed operating procedures.
  - The Theodolite operator tracks the aircraft using fine adjustment knobs while looking through the eyepiece.
  - The TLS technician can assist the theodolite operator by verbally guiding the operator to position the objective lens to acquire the aircraft as being viewed through the theodolite eyepiece.
12. Once the aircraft is in view, the theodolite operator will place the aircraft within the crosshairs of the eyepiece lens.
  13. The Theodolite operator will track the aircraft while viewing through the eyepiece lens within the crosshairs as follows:
    - a. Use the theodolite's vertical and horizontal fine adjustment knobs to maintain the aircraft in the crosshairs.
  14. When the theodolite operator reports over the radio that they are visually tracking the aircraft with the theodolite, press the "Start Record" button on the FTM pop up dialog.

**NOTE**

If the theodolite operator has not achieved visual track prior to the aircraft reaching about 2 degrees elevation or 6 NM, abort the flight approach and restart the approach.

15. Let the FTM system track the aircraft until the theodolite is out of range adjustment or the aircraft is 1 NM from the runway threshold.
16. Click on the "Stop Recording" button in the FTM window (See Figure 2-4).
17. In the Test Mode window, click on the "Abort" button (See Figure 2-4).
18. Inform the pilot that the profile has been completed and to continue to the next profile.
19. Inform the theodolite operator that the flight has been discontinued and to re-center equipment.

**NOTE**

- Do not change the default filename.
  - The calibration software in the steps below uses the default file naming convention.
20. In the FTM window, click on the "File" menu button to save the file:
    - a. Choose the "Close" option to close the recording file.
    - b. Make sure the .dif file is saved to the directory \\base2\c\run\raw.
    - c. Click on "Yes" to save the recorded file. Do not change the default filename. The calibration software in the following steps uses the default file naming convention.
  21. Exit the Test mode window by clicking on the "Exit Test mode" button.
  22. Repeat steps 3 through 17 a second time or until there are a minimum of two acceptable recorded, approaches that can be used to create the Calibration file.

**NOTE**

- If there are problems with any profile, repeat that profile so that the proper number of successful data sets is recorded.
  - Add detailed annotation to the notes section of the data sheet.
23. Record validation that profiles were completed, and the raw file numbers for the successful calibration profiles.
  24. Power on GTUs.

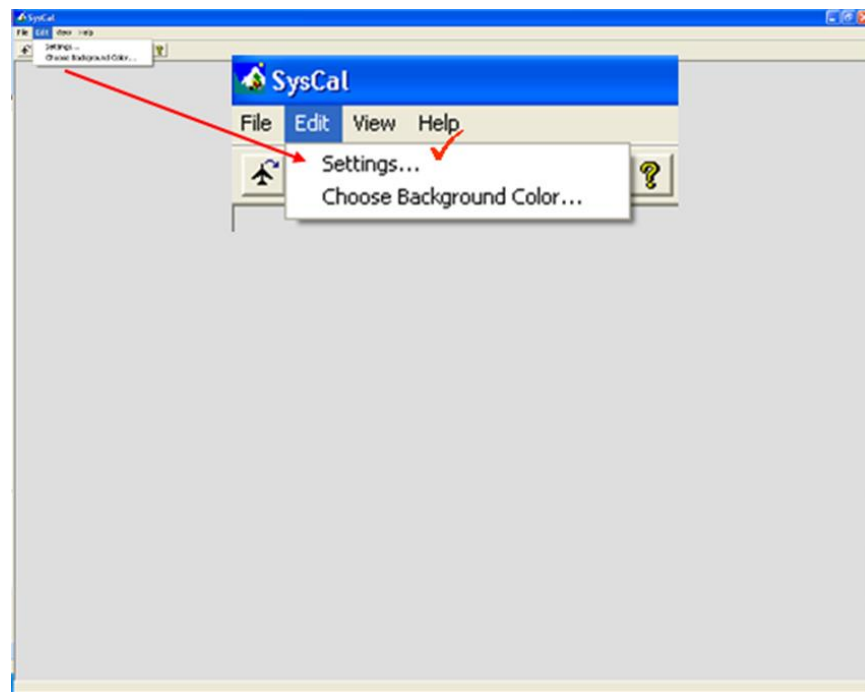
25. Click "Close" on the MI Test Mode window and wait for system to display "Track" in the status window.
26. On the MI, stop the system software.

## Calibration File Generation

### NOTE

The objective of this step is to generate the calibration file that provides precise localizer and glide slope alignment.

1. Ensure system software is stopped
2. From the MI, tools drop down menu, select SysCal Tool.
3. Under the "Edit" menu, open the "Settings" dialog (See Figure 2-5).



**Figure 2-5. SysCal Screen**

4. Select the "Calibrate with theodolite data" processing option (See Figure 2-6).

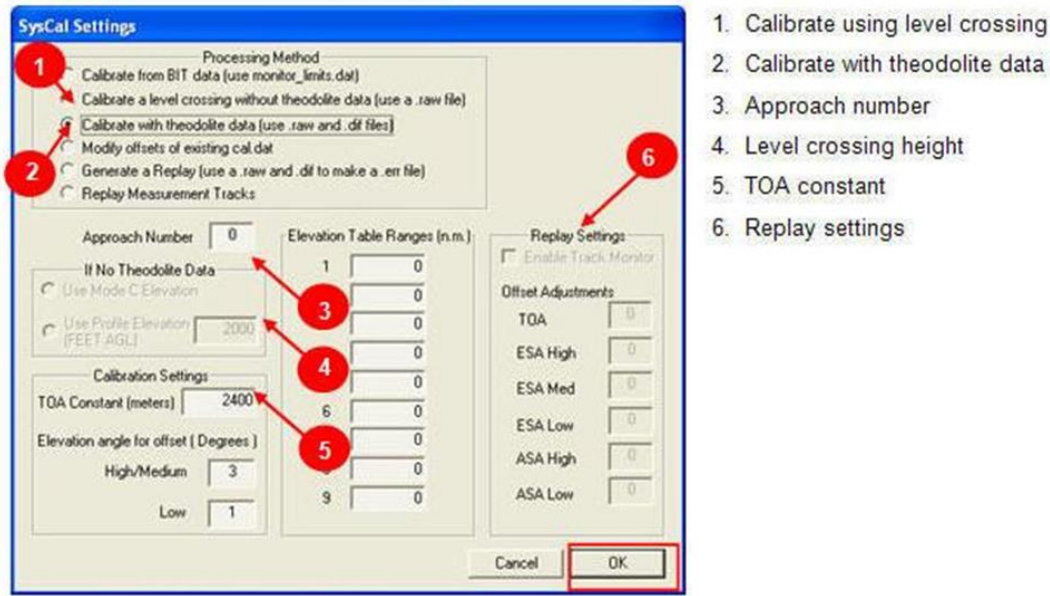


Figure 2-6. SysCal Setting Screen

#### NOTE

- Viewed on the RCU, counted top to bottom from the approach dropdown menu.
- The number is equal to the row of the approach from the approach selection box minus one.

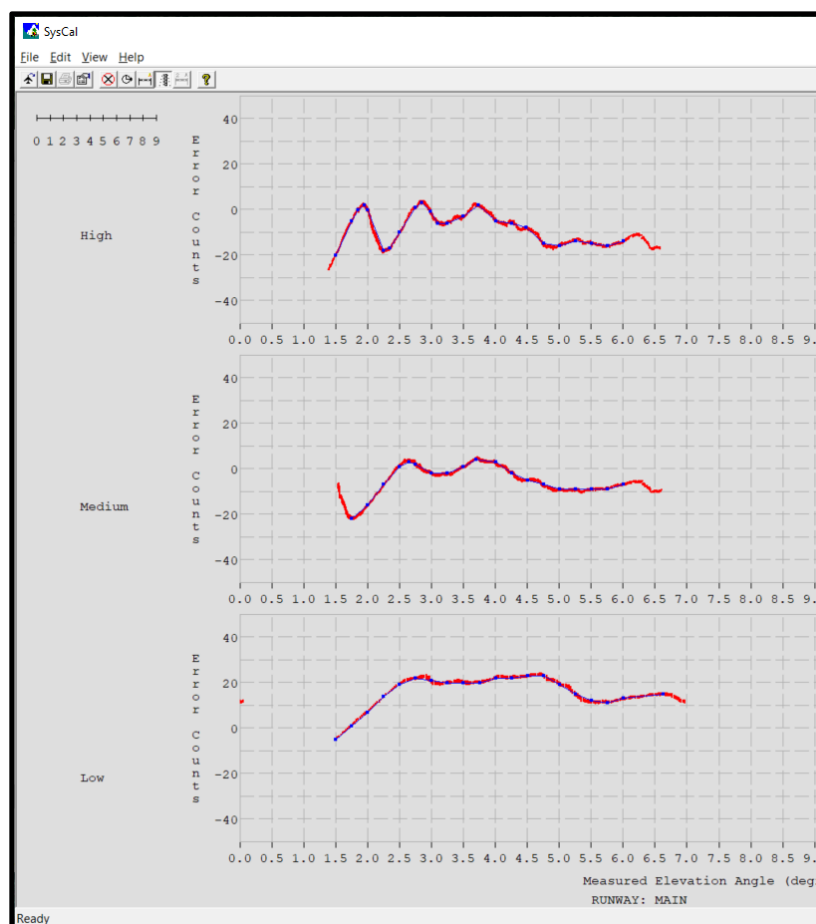
5. Enter the selected approach number used for the calibration approach.
6. Enter the calculated "TOA Constant". Use the default value 2600.
7. Click on "OK" to save and close the Settings dialog (See Figure 2-7).
8. Click the airplane button (See Figure 2-7).

#### NOTE

- A file selection box will be displayed.
  - The files (.raw) used to calibrate are selected from the "\\Base2\\c\\Run\\Raw" directory.
9. Using the flight log to decide, select the files that captured data for the level glide slope crossings.

#### NOTE

You can select multiple files using the shift and control keys.



**Figure 2-7. SysCal Calibration curves**

**NOTE**

- When the progress bar is done the calibration data for the file is initialized based on the data shown.
  - The blue line with dots displays the best fit for the data and is called the calibration curve.
  - Each dot on this line is a value that gets stored in the calibration file.
10. Drag the blue dots to improve data fit.

**NOTE**

- The goal is to represent the red data points with the blue curve.
  - Ignore obvious outlying data points or points that are on a secondary angle-of-arrival.
11. Make any necessary adjustments in the ESA plot as follows:
- a. Left click and hold down on the blue curve points.
  - b. Drag and place Blue to Red dots

- c. Move the points to best fit the red plot.
- d. Delete unnecessary points by pressing shift + left click on the point.
- e. Right click where you want a point inserted to add a point.

#### **NOTE**

- Typically, the tool generates a calculated curve. This curve can be used to examine the performance of the current flight data through replay runs.
  - A replay will use the system software along with the new calibration data to rerun the tracking solution for a flight with an aircraft under guidance.
  - Do not change the default filename because the calibration software in the steps below uses the default file name convention.
12. Save the new calibration file by using either the 'save', or 'save as...' command under the file menu, as a "cal.dat file.
  13. Record the validation of successful calibration file generation in the facility logbook.

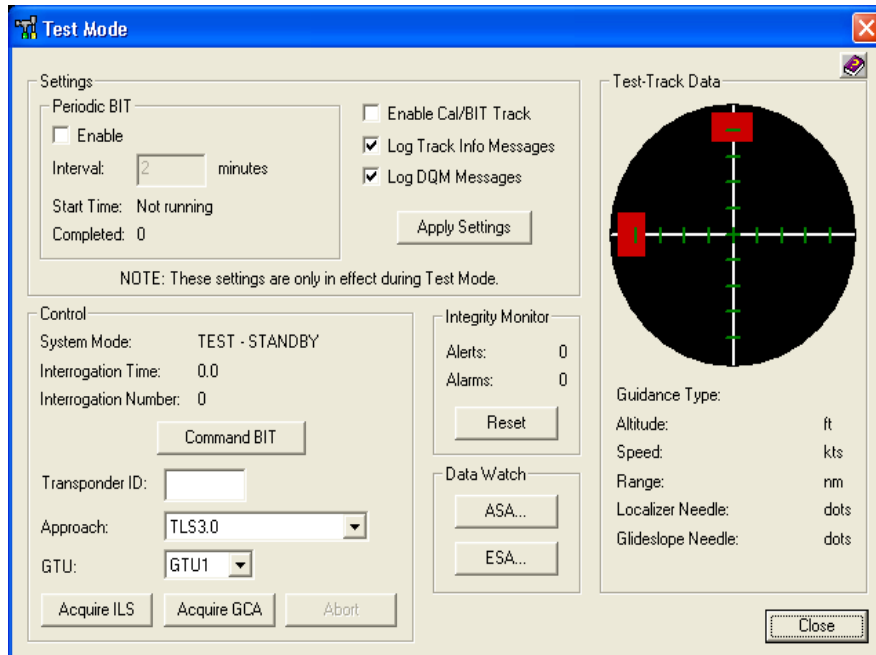
### **Validation Profile Approach**

1. Start system software.
2. If not already done, then in the FTM window, click on the "L" button to open a new FTM document.
3. Enter the preflight information.

#### **NOTE**

The preflight information can include specifics about the weather, aircraft, pilot, type, and purpose of flight, etc.

4. Click "OK."
5. Select applicable approach for profile from RCU dropdown labeled approach (See Figure 2-8).
6. Enter transponder ID on RCU.
7. Click acquire when aircraft is inbound.
8. Verify that TLS acquires a track on the aircraft.



**Figure 2-8. Test Mode Window**

9. Verify "TRACK LOC" is displayed for Guidance Status in the RCU window.
10. The TLS technician can assist the theodolite operator by verbally guiding the operator to position the objective lens to acquire the aircraft as being viewed through the theodolite eyepiece.
11. Once the aircraft is in view, the theodolite operator will place the aircraft within the crosshairs of the eyepiece lens, and alert TLS operator that aircraft is being tracked.
12. Select Record on FTM.

#### **NOTE**

If the theodolite operator has not achieved visual track prior to the aircraft reaching about 2 degrees elevation or 6 NM, abort the flight approach and restart the approach.

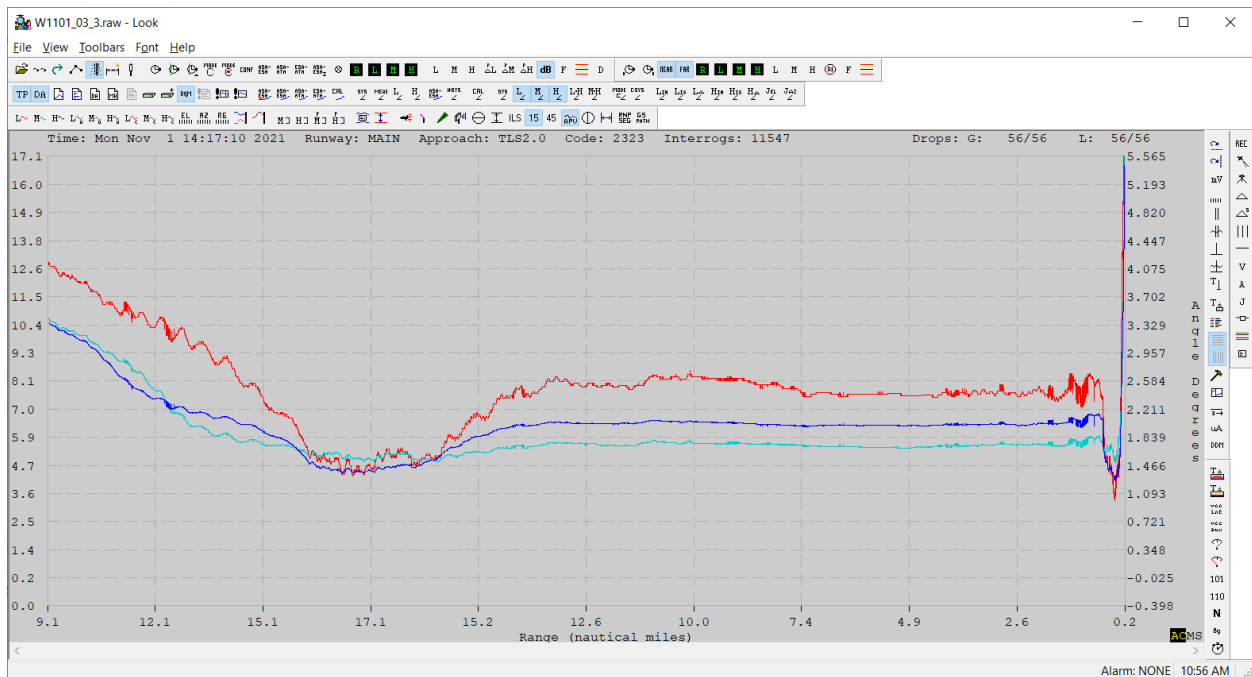
13. FTM operator will track the aircraft until crossing the threshold.
14. Click on the "Stop Recording" button in the FTM window.
15. Inform the theodolite operator that the flight has been discontinued and to re-center equipment.
16. Verify data is inside limit box on FTM window.
17. Communicate with pilot for feedback on approach and record in CTF.

## Calibration Validation Verification (Look Tool)

### NOTE

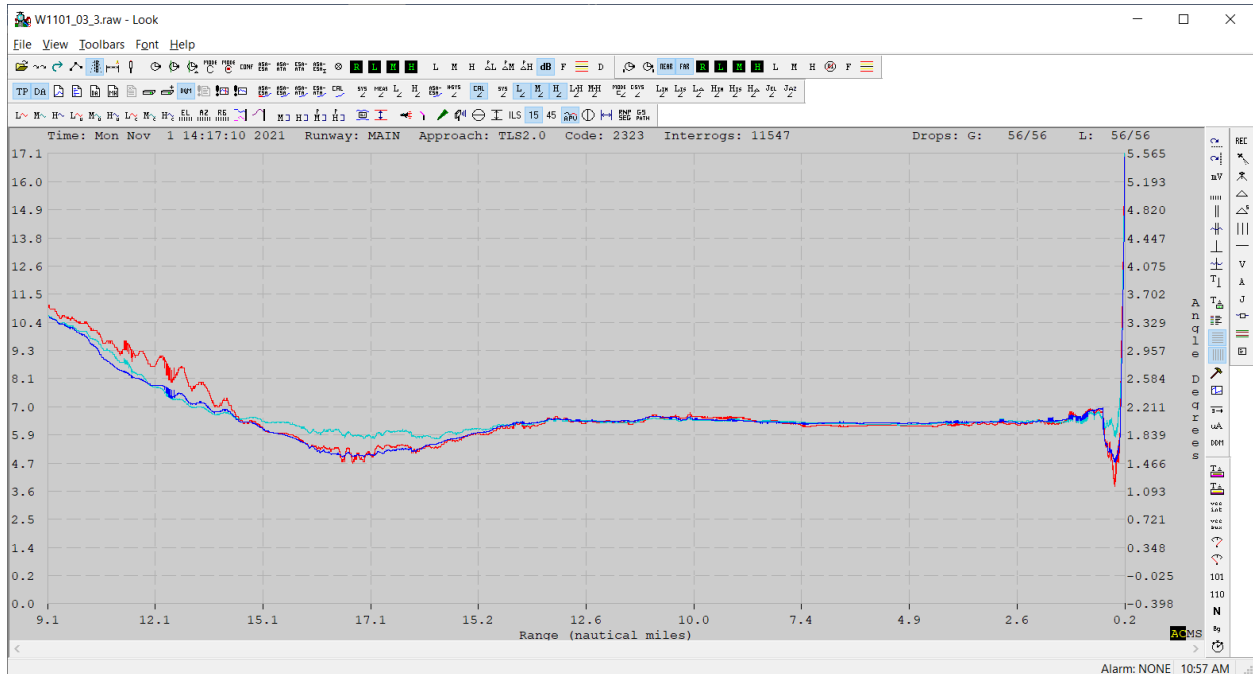
Use the Look tool as a quality control to verify that the calibration procedure has successfully aligned the calibration measurements.

1. From the MI start the Look tool (Look.exe) located in the Tools dropdown menu.
2. Use the tool to open the raw file on Base2 that was used for calibration.
3. Select measurement track buttons for the low, medium, and high-resolution measurements from the Track Processing Toolbar (See Figure 2-10).



**Figure 2-10. Glide Slope Measurement Track**

4. Select the Calibration Effect Toggle Button (See Figure 2-11) that allows viewing the effect of calibration.



**Figure 2-11. Calibration Effect**

5. Verify that the measurements are within the following tolerances of each other:
  - a. The high and medium are within  $\pm 0.75$  degrees.
  - b. The low is within 1.0 degree of both the high and medium.
6. If any of the measurement tracks need adjustment, use the SysCal tool to repeat steps 1-12.

## Alternative Calibration Procedure

### NOTE

- This calibration method uses a single aircraft flight profile and does not require a technician to visually track the aircraft with the theodolite.
- This procedure may be used as an alternative to the above procedure with theodolite if theodolite operations cannot be performed.
- The quality of this calibration process relies on the pilot to fly and maintain a specific course and altitude for the entire profile.
- Instead of calibrating the system to an independent truth source (the theodolite), the system will estimate the position of the aircraft based on the expectation that the pilot will fly straight.
- Final adjustment and independent verification of the localizer and glide slope parameters must be accomplished by Flight Inspection prior to commissioning and operational use.
- Since no theodolite is used for the initial alignment verification, it is likely that more adjustment will be needed during flight inspection.

1. To perform this procedure, repeat the above procedure used for the theodolite but in step 4 of section "CALIBRATION FILE GENERATION", select "calibrate using a level crossing without theodolite".
2. Select "using Mode C" or enter the predicted aircraft elevation in height above ground.

### Post Calibration Quality Checks

1. Click on the "SYSTEM SETUP" button on the Maintenance Interface.
2. Click on the "SETTINGS" button. Verify the following ANPC default settings and change as per user specification:
  - a. Data Recording
    - <Record Raw Files: Checked>
    - <Record GTU Files: Not-Checked>
    - <Record BIT RAM Files: Not-Checked>
  - b. Service Volume
    - <Azimuth Volume Edge:  $\pm 40^\circ$ >
    - <Elevation Volume Lower Edge:  $0.6^\circ$ >
    - <Elevation Volume Upper Edge:  $7^\circ$ >
    - <Localizer Range: 25 NM  $\pm 42^\circ$ >
    - <Glide slope Azimuth Volume Edge:  $12^\circ$ >
    - <Glide slope Range: 12NM>
  - c. Diagnostic Mode BIT Schedule
    - <Retry Interval: 3>
  - d. Interrogation
    - <Guidance: 22Hz>
    - <Surveillance: 2Hz>
3. Record verification that settings have been updated.

**END OF TASK**

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